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M 404 WASTE WATER TREATMENT SYSTEM

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with part 4, Subsection 1.2. "Summary of Work".

The waste water from Block 2, 3 & 4 shall be connected to a new Waste Water Treatment System. The bidder shall provide one (1) new Waste Water Treatment System for block 2, 3 & 4. New Waste Water treatment Plant shall have a capacity covering three (3) Blocks.

The drainage and waste water treatment equipment shall include but not be limited to the following major components with complete accessories.

1.1 Main Buildings Mechanical Drainage System

- Turbine room sump pump and motor
- Condenser pit sump pump and motor
- Turbine oil sump pump and motor
- HRSG pit sump pump and motor
- Drainage piping
- Pit level controller
- Control panel

1.2 Unit Neutralizing Pit Equipment for Combined Cycle Units

- Pit pump and motor
- Agitating air blower and motor
- Piping and valves (incl. agitating air piping and valves)
- Instruments
- Level controller
- Control panel

1.3 Waste Water Storage Pond Equipment

- Pond pump and motors

- Agitating air blower and motor
- Piping and valves (incl. agitating air piping and valves)
- Level controller
- Instruments

1.4 pH Control oxidation pit and mixing pit equipment

- Automatic pH controller
- Mixing pit agitator and motor
- Level controller
- Instruments
- Piping and valves

1.5 Coagulation-sedimentation Tank Equipment

- Agitator and motor
- Sludge pump and motor
- Piping and valves
- Instruments

1.6 Clear Water Pit Equipment

- Filter pit pumps and motors
- Level controller
- Piping and valves
- Instruments

1.7 Filter equipment

- Filter
- Filter air scrubbing agitator blower and motor
- Piping and valves
- Instruments

1.8 Neutralizing Pit equipment

- Pump and motor

- Agitator and motor
- Automatic pH controller
- Level controller
- Piping and valves
- Instruments

1.9 Purified Waste Water Pit Equipment

- Filter backwashing pump and motor
- Level controller
- Piping and valves
- Instruments

1.10 Sludge Enrichment Tank equipment

- Agitator and motor
- Sludge pump and motor
- Piping and valves
- Instruments

1.11 Sludge Storage Pond Equipment

- Discharge pump and motor
- Level controller
- Piping and valves
- Instruments

1.12 Chemical Injection equipment

- Dil. HCl Tank incl. agitator and motor
- Dil. HCl dosing tank and motors
- HCl gas seal tank
- Dil. NaOH tank incl. agitation and motor
- Dil. NaOH dosing pumps and motors
- Coagulant tank

- Coagulant dosing tank and motor
- Coagulant aid tank incl. agitator and motor
- Coagulant aid dosing pumps and motors
- Conc. HCl transfer pump and motor
- Conc. NaOH transfer pump and motor
- Chemical storage yard pit pump and motor
- Conc. HCl tank and conc. NaOH tank
- Level controllers
- Piping and valves
- Instruments

1.13 Control Unit System

- pH monitor and controller
- Flow integrating meter
- Control panel (incl. graphic panel)
- Instruments

1.14 Lighting

Although not clearly specified herein above any embedded materials for foundation, base plates, grouting materials, supports, electric motors, accessory piping, valves, instruments and blower silencers, concrete tank, equipment foundation, roof of concrete tank, access road and control room for the waste water treatment equipment as well as the piping rack foundation related to the waste water treatment equipment shall be included.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The codes and standards of the following organizations extracted from Part 4, Subsection 2.5, are specifically applicable to the design, manufacture and testing of the equipment included in this specification:

AGMA	American Gear Manufacturer Association
AWWA	American Water Works Association
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
NEMA	National Electrical Manufacturers Association
SSPC	Structural Steel painting Council
APHA	Americans Public Health Association
SNI	Standard Nasional Indonesia

3. DESIGN AND OPERATING CONDITIONS

The drainage and waste water treatment equipment to be procured and installed in accordance with these Specifications are the equipment to completely treat drainage and waste water discharged from Muara Tawar Combined Cycle Plant (Block 2,3&4) Add-On Project and discharged the treated waste water into the sea. Consequently, this equipment constitutes one of the most important equipment in view of environmental pollution control. Therefore, the equipment shall be highly reliable and in no way shall the performance be deteriorated through years of operation.

Installation and trial operation of the Waste Water Treatment System shall be completed before the date when Steam Turbine operate and the plant shall be ready for operation at the time.

The following drainage and waste water shall be treated as required.

- **Regularly discharged waste water treatment process**

Waste water to be treated by this equipment is as follows:

- Waste water from Turbine floor.
- Waste water from HRSG blow down (HRSG floor).
- Waste water from other machineries.

The waste water discharge shall be sent and stored in the waste water storage pond. Here, the waste water is agitated according to air atomized from the air diffuses pipe (tube) to make the water uniform. Then, it is sent to the pH control and oxidation pit. After its pH value has been regulated to an optimum level for coagulation and sedimentation, the coagulant and coagulant aid are injected into the waste water. Then, the waste water is sent to the mixing pit, coagulation - sedimentation tank where suspended solid is removed.

The clear water that has flowed out of the coagulation - sedimentation tank is collected in the clear water pit, then it is pumped into the filter and finally clarified. After pH value has been regulated in the neutralizing pit to the level permissible to discharge it into the sea, the water is discharged into the sea through the purified waste water pit.

However, should the pH value of the clear waste water exceed the specified level, discharge of waste water into the sea will be suspended and such water will be returned to the storage pond and treated again.

The suspended solid removed in the coagulation - sedimentation tank is collected and concentrated in the sludge enrichment tank. Then, it is sent to the sludge storage pond and further concentrated. After undergoing this process, the suspended solid is transported to the specified space on a tank lorry car and disposed of.

The filter is automatically washed by periodically returning clear water. The water used for filter washing is returned to and treated again in the storage pond.

- **Irregular waste water treatment process.**

The water to be treated in this process are as follows:

- Waste water from HRSG blow out (HRSG floor)
- Waste water discharged during HRSG chemical cleaning
- Fuel oil pump room drain from residual oil service tank

The waste water having irregular discharge such as HRSG chemical cleaning shall be required to be treated separately.

The reason is that the waste discharged during chemical washing contains a high concentration of Chemical Oxidation Demand (COD) contents. Therefore, such water is required to undergo oxidative destruction in the waste water storage pond prior to treatment in the waste water treatment equipment.

Treatment of waste water discharged on the occasion of blow-out and chemical cleaning of HRSG.

After the water has undergone oxidative destruction by injecting C.O.D. reduction agents such as H_2O_2 or $NaOCl$ in the waste water storage pond, and sent to the pH control and oxidation pit, this water is treated according to the same process as that for the regularly discharged waste water treatment.

Treatment of discharged waste water other than that discharged during blow-out and irregularly chemical cleaning of HRSG.

Since this discharged water contains a number of suspended solid contents mainly consisting of iron (Fe), it is required to control the pH value into an optimum level for removal of Fe by sedimentation.

Therefore, the Fe is required to be thoroughly removed through oxidation by aeration in the waste water storage pond. Then, this water is sent to the pH control and oxidation pit and treated according to the same process as that for the regularly discharged waste water treatment.

The Bidder shall provide all chemical supply to treat all commissioning waste water and shall quote other chemicals which are equal, in property to the chemical specified in the specifications.

The operation of waste water system shall be possible maintaining the treated water quality (pH and Suspended Solid) even if chemicals are not available.

Quality of treated water based on “*Peraturan Menteri Negara Lingkungan Hidup Nomor 08 Tahun 2009 tentang Baku Mutu Air Limbah bagi Usaha dan/atau Kegiatan Pembangkit Listrik Tenaga Termal*”

4. DETAILED REQUIREMENTS

4.1 Main Building Mechanical Drainage System

Turbine Room and HRSG Pit Sump Pump and Motor

Turbine room HRSG pit sump pump shall be vertical shaft, non clogging type. The motor shall be indoor and vertical type.

Condenser Pit Sump Pump and Motor

The condenser pit sump pump and motor shall be submerged type.

Turbine Oil Sump Pump and Motor

Shall be vertical shaft, non clogging. The motor shall be indoor and vertical type.

Machine Drainage and Floor Drain Piping

Place of installation shall be under floor. The pipe material shall be Carbon Steel pipe with asphalt jute cloth. The pipe diameter should not be less than 80 mm.

Sump Pump

- (a) The material of sump pumps shall be cast iron
- (b) Stop valves, check valves and pressure gauges shall be provided at the delivery of each pump.
- (c) The foundation bolts and nuts of pump shall be supplied.
- (d) Automatic lubricator (magnet type) shall be provided on the pumps.

- (e) Pumps shall be supplied and installed in accordance with Part 4, Specification M-008 Vertical Sump Pumps.

Level Control

- (a) Float type dial level gauge shall be installed at each sump pit.
- (b) Sump pump shall be automatically started and shut-down by the level controller in the pit.
- (c) Sump pit high level shall be alarmed at BTG board in the central control room.

Drainage Piping

- (a) Drain funnel for equipment and floor drain shall be provided, and the drain shall be led to the sump pit.
- (b) Drainage piping shall be installed under floor level between mat foundation and floor concrete slab, and outside surface of the pipe shall be covered with asphalt jute cloth.
- (c) Overflow piping from the turbine oil sump pit and turbine room sump pit to the condenser sump pit shall be provided, and the piping shall be covered with asphalt jute cloth.
- (d) The discharge of sump pumps shall be led to the unit neutralizing pit. The pipes shall be installed above the floor level.
- (e) The drainage piping for the HRSG, Stack, Bypass Stack shall be directly led to the unit neutralizing pit. The inside surface of the above piping shall be provided with acid proof lining, and outside surface of the buried piping shall be covered with asphalt jute cloth.

4.2 Unit Neutralizing Equipment

One (1) unit Neutralizing System shall be installed. The neutralizing capacity shall be based on the water amount needed to wash HRSG.

One unit neutralizing pit shall be underground reinforced concrete with acid and alkali proof coating type. Two (2) pit pumps shall be outdoor and vertical

shaft type. The material shall be of stainless steel or equivalent. The motors shall be outdoor and vertical type.

Air blower for the air agitators shall be installed on the pit.

Air agitating piping shall be installed in the pit.

The piping materials shall be acid proof.

Float type level dial indicator shall be equipped on the pit.

The pump shall be started up and shutdown automatically in accordance with the water level in the pit.

An outdoor type control panel shall be installed near the pit. Control switches, lamps, and alarms shall be provided on the panel. A common trouble alarm shall be provided in the central control room.

The foundation bolts and nuts of pump, blower, and panel shall be provided.

The size of inlet connection from the air blower shall be a minimum of 200 mm diameter.

A receptacle of waste water to be discharged during HRSG chemical cleaning shall be provided at the unit neutralizing pit pump outlet piping.

4.3 Waste Water Storage Pond Equipment

This pond is used to regular and irregular waste water to be separate treated. In this pond, an agitator shall be installed in order to prevent sedimentation of the SS (Suspended Solid) contents contained in the waste water to be treated and to agitate waste water by ejecting air from the air diffuses tube on the bottom of the pond to make uniform the quality of waste water in the pond.

The pump shall be started up and shutdown automatically in accordance with the water level in pond.

Float type level dial indicator shall be furnished for the pond.

All alarms shall be indicated on the board in the waste water treatment equipment control room.

The foundation bolts and nuts of pumps shall be supplied by the Bidder.

Materials for the air spray piping for agitator shall be acid-proof and alkali-proof.

The blower for air agitation shall be installed in the water storage pond. The air spray pipe for the agitator shall also be laid in the waste storage pond and the pH control and oxidation pit.

The waste water storage pond shall be semi underground reinforced concrete with acid and alkali-proof coating type. Pump shall be outdoor and vertical shaft type. The motor shall be outdoor and vertical type.

Agitating blower shall be rotary type with silencer for unit neutralizing pit, waste water storage pond and pH control oxidation pit.

4.4 pH Control Oxidation Pit and Mixing Pit Equipment

pH Control and Oxidation Pit

This pit is provided in order to let metallic hydroxides mainly consisting of $\text{Fe}(\text{OH})_3$ settle through control of pH of waste water to be treated and, at the same time, control the pH value into an optimum range for the coagulation sedimentation process.

- (a) The blower for air agitation to be used in this pit shall also be used as the blower for the waste water storage pond.

The air spray pipe to be attached in this pit shall be so arranged as to ensure uniform agitation of air.

- (b) pH control shall be made by injecting dilute HCl and dilute NaOH. The optimum pH shall be controlled between 6.0 to 9.0 and shall change automatically in conformity with the volume and quality of waste water, and type of coagulant.
- (c) pH control and oxidation equipment that comes into contact with liquid (waste water) shall be made of materials suitable for the quality of waste water and of shall be corrosion proof-finished.

Mixing Pit

This pit is installed in order to ensure easy and sufficient sedimentation and separation of the SS contents contained in the waste water to be treated. By

quantitatively injecting the coagulant and coagulant aid, large flocs are formed and the SS contents are captured in the flocs. Oil and other contents are included in the flocs, thereby it will be possible to sediment and separate the oil and other contents.

The coagulant (and coagulant aid) shall be injected into the pH control and oxidation pit.

The surface of the mixing pit that comes into contact with waste water shall be made of materials suitable for the quality of waste water and shall be corrosion proof finished.

pH Control Oxidation Pit

pH control oxidation pit shall be semi-underground reinforced concrete with acid and alkali-proof coating type.

Mixing Pit Agitation and Motor

Mixing agitation shall be paddle type. The motor shall be outdoor and vertical type.

4.5 Coagulation Sedimentation Tank Equipment

This tank is installed to sediment and remove the flocs formed in the mixing pit through separation according to gravitational force.

In this tank, solid and liquid are separated and the clear water is introduced into the clear water pit using gravity. The sedimented flocs are collected into the center of the tank bottom and sent to the sludge enrichment tank.

All Suspended Solid contents shall be sedimented sufficiently in the coagulation sedimentation tank, and the SS contents in supernatant liquid shall be made less than 30 ppm.

The discharge capacity of sludge shall be sufficient for treatment of the planned volume of waste water, even at the maximum load in view of the quality of waste water. The sludge pump shall be operated automatically by setting a timer.

The surface of this tank that comes into contact with waste water shall be made of the materials suitable for the quality of waste water or shall be corrosion proof -finished.

Coagulation sedimentation tank shall be semi-underground reinforced concrete Centre drive type and up-flow rate of 0.7 m/h or less. The driving unit shall be motor with reduction gear and turntable.

Sludge pump and motor shall be outdoor slurry horizontal type.

4.6 Clear Water Pit Equipment

This pit shall be installed to introduce waste water flowing down using gravity from the coagulation - sedimentation tank into the filter.

A clear feed pump shall be installed in this pit. This pump shall be started and stopped in conformity with the water level of the pit and the differential pressure of the filter.

Pit shall be semi-underground reinforced concrete type. Two (2) filter feed pumps shall be outdoor and vertical shaft type. The motor shall be outdoor and vertical type.

4.7 Filter Equipment

This filter equipment shall be installed to improve the quality of treated waste by making fine suspended solid contents being carried over from the coagulation - sedimentation tank pass through the filter medium (sand + anthracite coal) in order to remove the suspended solid contents.

Backwashing of this filter shall be carried out by using water and air. Backwashing of this filter shall be carried out once a day by means of timer as well as through detection of differential pressure, and the waste water discharged through backwashing shall be rooted to the waste water storage pond.

The air scrubbing blower shall be installed in the waste water treatment equipment control room.

Two (2) units filters shall be vertical pressure vertical type, for sand and anthracite.

Washing and cleaning shall be by water back-washing.

Air scrubbing blower shall be rotary type with silencer.

4.8 Neutralizing Pit Equipment

This pit equipment shall be installed to control the pH value of treated waste water from the filter to the pH value applicable to the standards for waste water to be discharged into the sea.

pH control in this pit shall be carried out automatically through interlocked operation of the pH gauge, and in the case where the pH value of treated water should not be within a range of 6.5 - 8.5, discharge of such water into the sea shall be suspended automatically and any such water shall be circulated (returned) to the waste water storage pond.

The pH value shall be indicated and recorded on the panel in the waste water treatment equipment control room.

The neutralizing agents to be used for pH control shall be 10% HCl and 10% NaOH

The surface of the equipment that comes into contact with the neutralizing agents shall be made of acid and alkali proof materials.

The pit pump shall automatically be started and stopped in conformity with the water level of the pit.

The flow integrating meter shall be attached at the pit pump outlet, and any indication and recording there from shall be made on the panel in the waste water treatment equipment control room.

4.9 Purified Waste Water Pit Equipment

This pit equipment shall be installed to check the quality of treated waste water flowing from the neutralizing pit prior to discharging into the sea.

This pit shall also be used to wash the filter by reversing clear water.

Oil separator shall be provided and used as this pit.

The filter backwashing pump to be installed in this pit shall automatically be started and stopped in conformity with the water level in this pit, or by means of timer or through transmission of signal of differential pressure of the filter.

4.10 Sludge Enrichment Tank Equipment

This tank shall be installed to enrich the sludge separated in the coagulation - sedimentation tank.

The sludge pump shall be so designed and constructed as to sufficiently and smoothly discharge the sludge enriched in the sludge enrichment tank into the sludge storage pond. The sludge pump shall be automatically operated through setting of timer.

Any water flowing over the sludge enrichment tank shall be recovered into the waste water storage pond.

4.11 Sludge Storage Pond Equipment

This pond shall be installed to further enrich through natural sedimentation and store the sludge discharged from the bottom of the sludge enrichment tank.

Any supernatant water in the sludge storage pond shall be routed to the waste water pond by means of the discharge pump.

The discharge pump shall automatically be started and stopped in conformity with the water level in the pit.

Pond shall be semi-underground reinforced concrete centre drive type. Up-flow rate shall be 0.5 m/hr or less.

Effect of Sludge Enrichment : average suspended solid concentration at inlet 1.5% and average suspended solid concentration at outlet 4.0%.

The driving unit shall be motor with reduction gear turn-table.

Sludge pump shall be outdoor and slurry horizontal type. The motor of pump shall be outdoor and horizontal type.

4.12 Chemical Injection Equipment

The chemical injection equipment shall be interlocked with the master switches, etc. for the pH control and oxidation pit equipment, the mixing pit equipment, and neutralizing pit equipment.

The quantity of chemicals to be injected shall be controlled by changing the stroke of the pump, and chemical injection shall be carried out continuously by using the control valve.

pH control shall be carried out automatically through interlocked operation of the pH gauge, and injection quantities of coagulant and coagulant aid shall be controlled by the controllers on control panel in the waste water treatment control room.

Pumps shall be supplied and installed in accordance with Part 4, Specification M-009 Metering Pumps.

Shop fabricated tanks shall be supplied and installed in accordance with Part 4, Specification M-012.

Operation of System

In order to ensure labor saving and safety as well as facilitate operation and control, automatic operation system shall be adopted as far as possible. In other words, the waste water treatment system shall be put into automatic operation if the control switches of the respective equipment have been set to AUTO and the master switch has been turned ON.

The operation and control of the waste water treatment system shall in principle be carried out according to the Waste Water Treatment System Monitor and Control Panel installed in the control room.

Moreover, the monitor and control instruments shall be furnished on the control and monitor panel, and the operating conditions of the respective equipment shall be displayed on the graphic panel in order to ensure easy operation and control of the waste water treatment system.

This equipment is installed to inject hydrochloric acid (including diluted hydrochloric acid) and caustic (including diluted caustic) into the pH control

and oxidation pit and the neutralizing pit in order to control the pH value of treated waste water into optimum level.

This equipment is also used to inject the coagulant and coagulant aid in order to ensure formation of large flocs through coagulation and smooth sedimentation of flocs.

Concentrated Hydrochloric Acid

The liquid state hydrochloric acid is received into the underground concentrated hydrochloric acid tank from a tank lorry using the gravity force. Tank shall be horizontal and cylindrical type. The density shall be 35% as HCl.

Concentrated NaOH Tank

The liquid state caustic soda is received into the underground concentrated caustic soda tank from a tank Lorry using gravity force.

Tank shall be vertical cylindrical FRP or equivalent type. The density shall be 45% as NaOH.

Diluted HCl Tank

The liquid state hydrochloric acid is pumped into the diluted hydrochloric acid tank from the concentrated hydrochloric acid tank to be diluted.

Preparation of 10% HCl solution shall be made automatically in conformity with the level of the diluted hydrochloric acid tank.

Tank shall be vertical cylindrical FRP or equivalent type. The density shall be 10% as HCl.

Tank Agitator and Motor shall be paddle type. The motor shall be outdoor and vertical type.

HCl Gas Seal Tank

Tank shall be vertical cylindrical natural rubber coating type.

Diluted HCl Dosing Pump shall be outdoor and horizontal diaphragm type. The motor shall be outdoor and horizontal type.

Diluted NaOH Tank

The liquid state caustic soda is pumped into the diluted caustic tank from the concentrated caustic tank. Preparation of 10% NaOH solution shall be made automatically in conformity with the level of the diluted caustic soda tank. Tank shall be vertical cylindrical FRP or equivalent type. The density shall be 10% as NaOH.

Tank Agitator and Motor shall be paddle type. The motor shall be outdoor and vertical type.

Diluted NaOH Dosing Pump and Motor. Two (2) pumps shall be provided and shall be outdoor horizontal diaphragm type with high chrome nickel material. The motor shall be outdoor and horizontal type.

Ferric chloride (liquid FeCl_3) shall be used as a coagulant. Tank shall be vertical cylindrical FRP or equivalent type. The density shall be 80% as $\text{FeCl}_3 \cdot 6 \text{H}_2\text{O}$.

Coagulant Dosing Pump and Motor. Two (2) pumps shall be outdoor and horizontal diaphragm type with natural rubber coating. The motors shall be outdoor and horizontal type.

The high molecular coagulant aid (powder state) shall be used as a coagulant aid. Dissolving, of this coagulant aid shall be made easily into a uniform concentration. Tank shall be vertical cylindrical FRP or equivalent type. The density shall be of 0.1 % as high molecular coagulant aid.

Control Mechanism

Pumps and similar equipment shall be controlled through ON/OFF operation of the level switches.

pH control of waste water shall be carried out through ON/OFF operation of control valves by signal from pH analyzer.

The instruments and control units to be used shall be either of an electrical or pneumatic type.

Alarm shall be issued should any trouble arise in the instruments or equipments or any process variable such as water level. pH value, etc. Any alarm to be issued shall be indicated in the central control room.

Coagulant aid tank agitator shall be paddle type the motor shall be outdoor and vertical type.

Coagulant aid dosing pumps shall be outdoor and horizontal diaphragm type with epoxy-resin coating the motor shall be outdoor and horizontal type.

Concentrated HCl transfer pump shall be outdoor and horizontal diaphragm type with natural rubber coating. The motor shall be outdoor and horizontal type Density shall be 35% as HCl.

Concentrated NaOH transfer pump shall be outdoor and horizontal diaphragm type with Epoxy-resin coating. The motor shall be outdoor and horizontal type Density shall be 45% as NaOH.

Chemical storage yard pit pump shall be submerged with acid and alkali proof coating type the motor shall be outdoor and submerged type.

4.13 Control Unit System

pH meter shall be glass electrode type. pH detector, pH indicator, pH recorder shall be provided three (3) sets each. Alarm setter shall be double pin type. pH metering range ($\text{pH} = 2 - 14$). pH detection, locations are pH control and oxidation pit and Neutralization pit.

Flow integrating shall be orifice type. Flow detector, flow indicator, flow integrating meter shall be provided, flow metering range. Flow detection location shall be provided on neutralizing pit pump.

Local control panel shall be provided. Wall mounted indoor type shall be used for Turbine room sump pump and motor, condenser pit sump pump and motor, turbine oil sump pump and motor. Unit neutralizing equipment shall be outdoor type.

Control Panel for The Waste Water Treatment Equipment

Type : Self-standing type made of steel plate
Number : One (1) set

- The control panel which shall consist of indicator, control switches, alarm indicator lamps and graphic panel shall be installed in the waste water treatment equipment control room. The control panel shall be designed and constructed to ensure easy centralized supervision and operation.
- The Bidder shall submit the drawing of graphic panel to the Owner for approval.

Control Panel for Unit Neutralizing Pit Equipment

Type : Self-standing waterproof type made of steel plate
Number : One (1) set

This control panel shall be installed outdoors in the vicinity of the unit neutralizing pit.

4.14 Piping

All portions of piping required to be washed, replaced in view of the construction or to be dismantled at the time of inspection of the equipment shall be connected with flanged joints.

The portions of the sludge line, etc., which might possibly be subjected to clogging shall be provided with washing water lines.

The internal surface of the piping which comes into contact with hydrochloric acid shall be lined with rubber, and the piping for caustic shall be made of stainless steel. All appropriate measures shall also be taken against possible leakage of hazardous chemicals.

The sludge line piping shall be of stainless steel.

The portions of the piping penetrating into the respective tanks shall be made of stainless steel, and the pipes to be laid for the agitators inside the tanks shall be PVC.

Piping shall be in accordance with Part 4, Technical Specifications M-001.

4.15 Valves

All valves shall be of anti corrosive material and shall be mounted at positions permitting convenient operation.

The valves to be installed outdoors shall be of such a construction as not to cause any malfunction.

The valves to be used for any acid or alkali chemical line shall be of such a construction that any glands will not come into direct contact with the external air.

Stop valves for the sludge line shall be the diaphragm type.

The nameplates indicating the Numbers and Names of the respective valves shall be attached to the respective valves.

The Valves shall be in accordance with Part 4, Technical Specifications M-004 and M-005.

4.16 Painting

The painting shall be in conformity with Part 4, Technical Specifications M-001, S-420 and S-430.

4.17 Scaffold, Handrail and Ladder

Platforms handrails and ladders shall be provided as required for operation and control, inspection and maintenance of the respective equipment while taking into account the safety of the above works.

4.18 Special Tools and Standard Tools Set

The Bidder shall provide the special tools and standard tool sets for the drainage and waste water treatment equipment.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 Testing and Guarantees and shall guarantee the following system guarantees :

5.1 Performance Tests

After equipment is put in service the Bidder will run a check test to ascertain whether the equipment is performing properly in every respect. Bidder shall arrange to have his authorized representative present to witness and assist in these tests.

5.2 Site Test

The piping will be subject to a field hydrostatic leak test of 1.5 times the design pressure after it is installed. All work-manship and materials shall be of such quality as to ensure that no defects due to faulty fabrication or faulty materials will develop during the testing.

5.3 Chemical Waste Water Effluent Guarantee

Bidder shall guarantee that the chemical waste water treatment system furnish by him, when installed, maintained and operated in accordance with his written instruction and recommendations will produce an effluent whose pH is not lower than 6.0 and not higher than 9.0.

Quality of treated water based on “*Peraturan Menteri Negara Lingkungan Hidup Nomor 08 Tahun 2009 tentang Baku Mutu Air Limbah bagi Usaha dan/atau Kegiatan Pembangkit Listrik Tenaga Termal*”

6. QUALITY ASSURANCE / QUALITY CONTROL (QA/QC)

Quality Assurance/Quality Control shall conform to the requirements of Part 4, Subsection 2.7.

The quality control procedures and documentation so specified in Clause 5.1 shall be included but not be limited to the following:

- Material certification.
- Welding
- Nondestructive testing
- Heat treating
- Packaging and shipping
- Shop Test procedures
- Field Test procedures

- Cleaning and Painting

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall furnish and install complete instrumentation and control systems and devices for the Waste Water Treatment System including but not be limited to those listed and describe in this Technical Specification.

7.2 Design Requirements

The design of instrumentation and control systems and devices shall be in accordance with Part 4, Technical Specification I-001.

Programmable Logic Controller (PLC) shall be used for controlling the Waste Water Treatment. PLC shall be in accordance with the requirements of Part 4, Technical specification I-005. The PLC shall be located in the cabinets located in Waste Water Treatment Building as applicable.

Control logic for the Waste Water Treatment Plant shall be designed for “Fail-Safe” operation.

The Bidder shall furnish and install a waste water control panel in waste water treatment control room. This shall provide centralized coordinated control for the waste water treatment system.

The centralized control shall include provision for automatic and manual control of all the process and equipment necessary for the proper functioning of the above systems. The panel shall contain all instruments and devices necessary for safe and efficient control such as indicators, recorders, totalizers, control stations, M/A stations, control switches, push buttons and indicator lights. Annunciator and mimic representations shall be provided in accordance with the requirements of Part 4, Technical Specification **I-002**.

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The design of local control panels shall be in accordance with Part 4, Technical Specification I-002.

On/off control, sequencing, interlocking and similar control functions for the Wastewater Treatment shall be carried out by a single programmable controller specified herein. The design of locally mounted instrument panels

shall be in accordance with Part 4, Technical Specification I-003. The complete operation of the equipment shall be fully monitored for automatic shutdown and alarm in case of malfunction. Failure of any operation to reach completion within a predetermined time interval shall be annunciated and shall stop the automatic process until satisfactory conditions have been re-established. Serial data link shall be provided to interface with Plant Distributed Central System DCS.

The control systems shall prevent the initiation or continuation of any automatic operation or sequence of operations which would conflict with any operation or sequence already in progress, whether such operation or sequence is under automatic or manual control.

The operating status of all waste water treatment equipment and of any automatic sequence which is in progress shall be indicated by waste water control panel in the waste water treatment building.

Means shall be provided for manual control from the control panels of every operation performed by the automatic control systems. Such provisions shall permit manual entry into any automatic sequence, repetition under manual control of any completed operation in the sequence and return to automatic control.

Signals transmitted from locally mounted transmitters to controllers, indicators, recorders etc. shall be 4-20 mA DC.

The PLC I/O for plant monitoring system (DCS) shall be provided for the waste water treatment system in accordance with Part 4, Technical Specification I-200. The PLC I/O list shall be subject to the Owner's review and approval.

The discharge of each pump, blower, compressor, fan etc. shall be provided with pressure indication.

8. ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Waste Water Treatment System. Equipment furnished shall include, but not be limited to, the followings:

1. Motors.
2. Power panels.
3. Conduits, cables and wiring.
4. All electrical control, monitoring and instrumentation devices, and other appurtenance required, shall be provided by the Bidder.

For detailed electrical requirements, see Part 4, Technical Specification E-210 for Low Voltage Motors.

The Bidder shall provide all special equipment required for variable speed devices, etc.

8.2 Power Panels

For detailed electrical requirements, see Part 4, Technical Specification E-001 and E-280.

The panel shall have a degree of protection IPW 55 of IEC 60529.

8.3 Conduits, Cable and Wiring

All motors and electrical, control, monitoring and instrumentation devices furnish and installed by the Bidder shall be wired by him to the power and local control panel respectively.

The wiring between those panels shall also be included.

The required power, control, instrumentation and thermocouple extension cables shall be provided by the Bidder.

For detailed electrical requirements, see Part 4, Technical Specification E-230, E-231 and E-232.

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